

Data Warehouse Design

Analytical Requirements

The Data Warehouse is designed to support the analysis of historical card transaction activity. It should answer the following business questions:

1. Which customer age groups generate the highest transaction value and number of transactions in different merchant categories?

Business value: This helps identify relevant customer segments for targeted offers, rewards, and marketing campaigns.

2. In which regions and merchant categories is card spending highest?

Business value: This helps identify strong markets and attractive merchant categories for promotions and potential partnerships.

3. How do transaction value and number of transactions change across months, quarters and years?

Business value: This reveals spending trends and seasonality and supports campaign planning and performance comparison over time.

4. Which card types generate the highest transaction value and usage, and how does their performance change over time?

Business value: This helps evaluate the performance and adoption of different card products and supports product development and card strategy.

Multidimensional ER Model (mER)

Fact: Transaction.

Grain: One fact row represents one individual card transaction.

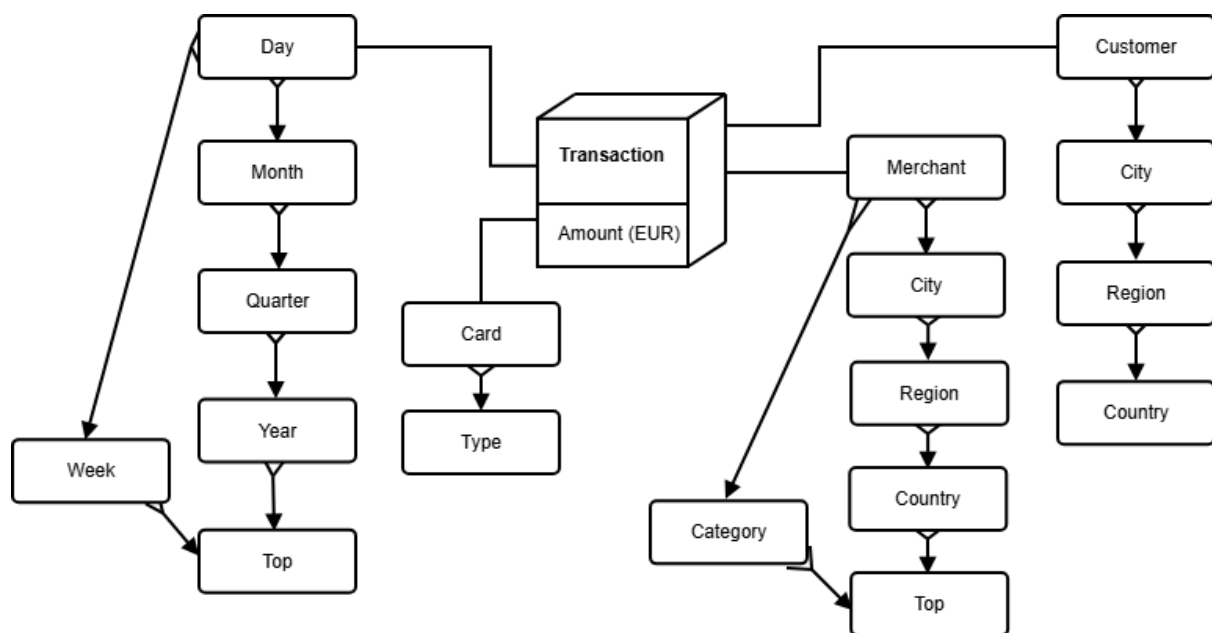
The Transaction fact is the central fact because the analytical requirements focus on transaction value and transaction volume across customers, merchants, card types and time.

Measures:

The Transaction fact contains the following measure:

- Amount (EUR): The monetary value of an individual card transaction, converted to EUR during the ETL process.

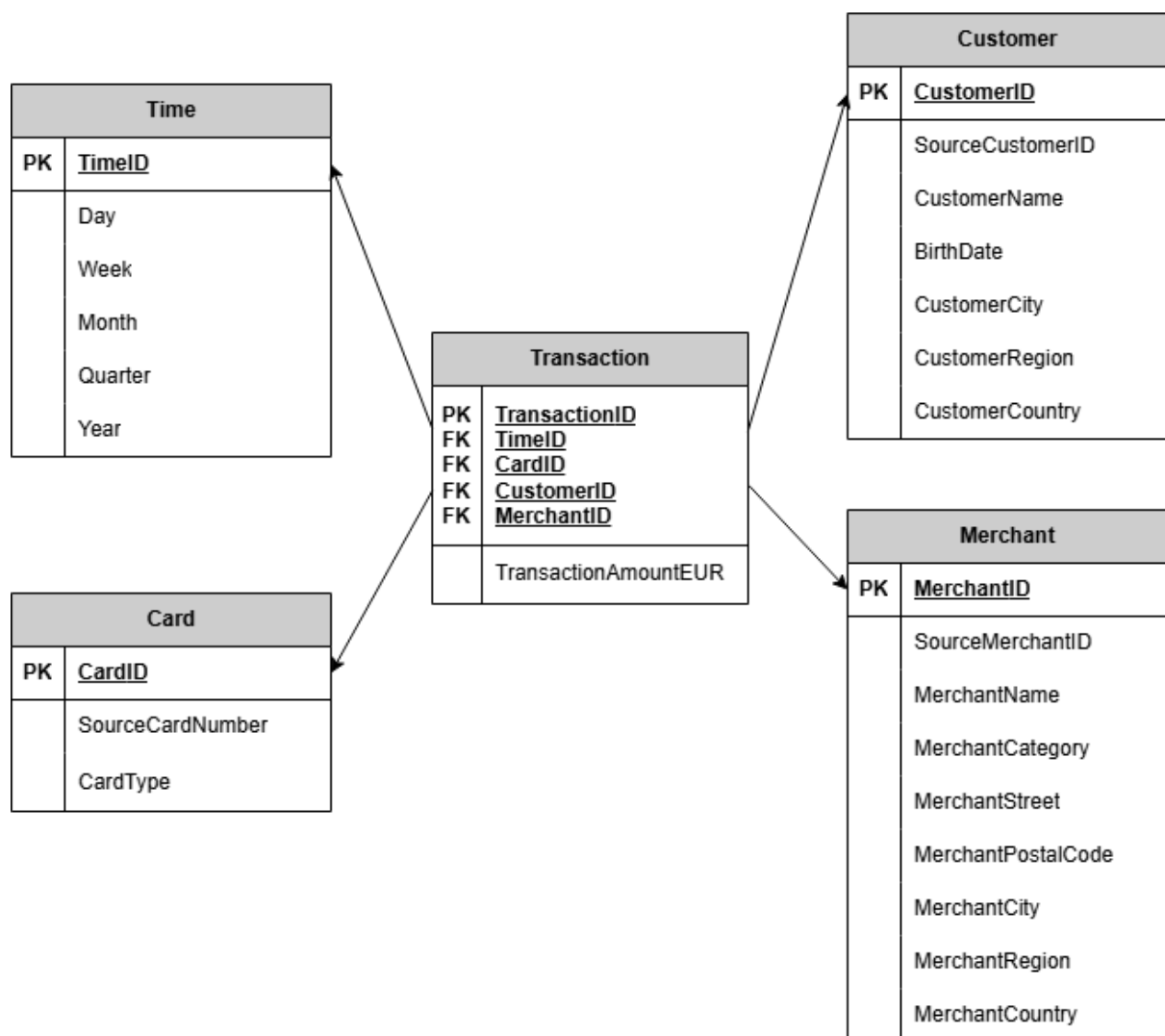
Based on the atomic transaction grain, analytical values such as total transaction value, number of transactions and average transaction value can be derived using `SUM(Amount)`, `COUNT(*)` and `AVG(Amount)`.



Logical Star Schema

The conceptual multidimensional model was transformed into a logical Star Schema. The central fact table is Transaction, with a grain of one individual card transaction. It is connected to the Time, Customer, Card, and Merchant dimensions through foreign keys. TransactionAmountEUR represents the monetary measure used for the analysis.

For the entity dimensions, source identifiers are retained alongside the Data Warehouse identifiers to preserve traceability to the source system.



Slowly Changing Dimensions (SCD)

Dimension: Time

SCD Type: Type 0

Time attributes are fixed and do not change after they have been created. Therefore, no historical tracking or overwriting is required.

Dimension: Customer

SCD Type: Type 0 + Type 1 + Type 2

Attribute	SCD Type	Reason
SourceCustomerID	0	Business identifier should not change
BirthDate	0	Fixed attribute; also a course example
CustomerName	1	Name corrections or changes overwrite the previous value; historical names are not analytically relevant
CustomerCity	2	Customer moves should remain historically traceable
CustomerRegion	2	Past transactions should remain assigned to the region valid at that time
CustomerCountry	2	Past transactions should remain assigned to the country valid at that time

Dimension: Merchant

SCD Type: Type 0 + Type 1 + Type 2

Attribute	SCD Type	Reason
SourceMerchantID	0	Business identifier should not change
MerchantName	1	Name corrections or rebranding can overwrite the previous value; historical merchant names are not required for our analysis
MerchantCategory	2	Past transactions should remain assigned to the merchant category valid at that time

MerchantStreet	2	Location changes should remain historically traceable
MerchantPostalCode	2	Past transactions should remain assigned to the location valid at that time
MerchantCity	2	Past transactions should remain assigned to the city valid at that time
MerchantRegion	2	Important for historical regional spending analysis
MerchantCountry	2	Past transactions should remain assigned to the country valid at that time

Dimension: Card

SCD Type: Type 0

Card attributes are treated as fixed for the lifetime of an individual card. Therefore, no historical tracking or overwriting is required. A different card type would normally be represented by a new/reissued card

=>

Dimension	SCD Type(s)
Time	Type 0
Customer	Type 0 + Type 1 + Type 2
Merchant	Type 0 + Type 1 + Type 2
Card	Type 0

Logical Star Schema including SCD

Customer and Merchant are historized using SCD Type 2 by adding ValidFrom and ValidTo. Their existing CustomerID and MerchantID already serve as the Data Warehouse keys for individual historical versions, so no additional SCD key (SCDID) is required. Time and Card remain structurally unchanged.

